THE WEIGHTED FIBRE-COUNT INTEGRAL OF A WEIGHTING ON A QUASI-COMPACT LOCALLY QUASI-FINITE MAP ATTAINS ITS MAXIMUM

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 22730-22735 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 22737-22746, the complete author proof (8 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Terminal small lemma of the weighting chapter-section, feeding the etale-descent weighting card collected this morning (L16088 weighting-check-after-etale-base-change).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Definition of weighting and of int_f w (chapter definitions); chapter lemma jumps-w (constructibility of weightings' jump loci); Properties, lemma stratification-locally-finite-constructible; Morphisms, lemma locally-quasi-finite-qc-source-universally-bounded.

Source reference. Original lemma 0F3G: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 22514–22518. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03JA: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 22746–22746. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0F2M: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 22746–22746. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to Y$ be a locally quasi-finite morphism with X quasi-compact. Let $w : X \to \mathbf{Z}$ be a weighting of f. Then $\int_f w$ attains its maximum.*

Proof. It follows from Lemma 0F3G and Properties, Lemma 0F2M that w only takes a finite number of values on X. It follows from Morphisms, Lemma 03JA that $X \to Y$ has bounded geometric fibres. This shows that $\int_f w$ is bounded. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.